

# Managing supply chain performance using a real time Microsoft Power BI dashboard by action design research (ADR) method

Dewan Hafiz Nabil, Md. Habibur Rahman, Altaf Hussain Chowdhury & Brenno Castrillon Menezes

**To cite this article:** Dewan Hafiz Nabil, Md. Habibur Rahman, Altaf Hussain Chowdhury & Brenno Castrillon Menezes (2023) Managing supply chain performance using a real time Microsoft Power BI dashboard by action design research (ADR) method, Cogent Engineering, 10:2, 2257924, DOI: [10.1080/23311916.2023.2257924](https://doi.org/10.1080/23311916.2023.2257924)

**To link to this article:** <https://doi.org/10.1080/23311916.2023.2257924>



© 2023 The Author(s). Published by Informa UK Limited, trading as Taylor & Francis Group.



Published online: 27 Sep 2023.



Submit your article to this journal [↗](#)



Article views: 13902



View related articles [↗](#)



View Crossmark data [↗](#)



Citing articles: 13 View citing articles [↗](#)



Received: 04 May 2023  
Accepted: 04 September 2023

\*Corresponding author: Md. Habibur Rahman, Department of Industrial Engineering and Management, Khulna University of Engineering & Technology, Khulna 9203, Bangladesh; Division of Engineering Management and Decision Sciences, Hamad Bin Khalifa University, Doha 5825, Qatar  
E-mail: [mdra47330@hbku.edu.qa](mailto:mdra47330@hbku.edu.qa)

Reviewing editor:  
D T Pham, School of Mechanical Engineering, University of Birmingham, UK

Additional information is available at the end of the article

## PRODUCTION & MANUFACTURING | RESEARCH ARTICLE

# Managing supply chain performance using a real time Microsoft Power BI dashboard by action design research (ADR) method

Dewan Hafiz Nabil<sup>1</sup>, Md. Habibur Rahman<sup>1,2\*</sup>, Altaf Hussain Chowdhury<sup>1</sup> and Brenno Castrillon Menezes<sup>2</sup>

**Abstract:** Supply chain (SC) performance is being advanced through digitalization, automation, and real-time visibility, leading to improved operational efficiency, optimized inventory management, and proactive solutions for problems. This study focuses on managing SC performance by developing a real-time Microsoft Power BI dashboard using the ADR method. The aim is to help small business owners effectively and affordably manage their supply chains. The research analyzed data from Ly Foods Ltd. and utilized Microsoft Power BI to create the dashboard. The study highlights how this state-of-the-art dashboard enhances the measurement of operational and decision-making processes by providing efficient, visible, accessible, and shareable information. The findings demonstrate the potential benefits of leveraging advanced technologies to enhance supply chain management (SCM) practices. This research makes a significant contribution to the field of SCM by illustrating how SC performance can be managed using the ADR method and the SC key performance indicators (KPIs). The results of this study have important implications for businesses of all sizes, as they enhance SC efficiency and profitability.

**Subjects:** Industrial Engineering & Manufacturing; Engineering Management; Production Engineering

**Keywords:** business intelligence; Microsoft Power BI; supply chain dashboard; supply chain performance

## 1. Introduction

In today's digital transformation age, within the progress from Industry 4.0 to Industry 5.0, the amount of data gathered by companies has increased dramatically due to the digitization or digitalization of their operational activities along with the evolution of new technologies such as the IoT (Internet of Things) and the widespread utilization of ERP (Enterprise resource planning) software (Trieu, 2017). This vast amount or volume of data has no or little value unless it can be accessed and processed effectively; therefore, the proper velocity of a variety of data to be measured, processed, and communicated is a fundamental variable to be reached for a viable dataset to be used in decision-making problems or as parameters for performance indicators. Business intelligence (BI) systems have become increasingly essential for companies to make educated judgments

about their operational procedures, as they efficiently arrange and transform data into knowledge that serves as the foundation for decision-making (Becker & Gould, 2019; Hannila et al., 2022; Iliashenko et al., 2019). The importance of measuring SC performance for the success of firms and customer satisfaction cannot be overstated. Effective SCM can lead to cost savings, improved customer service, and increased competitiveness. However, traditional SCM systems can often be outdated and challenging to use, leading to inefficient operations and suboptimal decision-making. With the aim of improving SC performance measurement, this research focuses on creating a dashboard using Microsoft Power BI. The dashboard's creation is based on the idea that a simpler, open, real-time, condensed, and collaborative SCM dashboard can help small business owners manage their SC more effectively and affordably (Iliashenko et al., 2019; Kuzmina et al., 2022). With its powerful analytics and reporting tools, Power BI allows SC managers to monitor, analyze, and optimize various aspects of the SC. By integrating data from multiple sources, including ERP systems, inventory management systems, and transportation management systems, Power BI enables real-time tracking of KPIs such as order fulfillment rates, inventory levels, lead times, and supplier performance. These visualizations help identify inefficiencies and areas for improvement, enabling timely decision-making and proactive actions (Becker & Gould, 2019; Brandão et al., 2016; Oey et al., 2021). Power BI, developed by Microsoft, is a highly acclaimed BI system known for its extensive array of interactive data visualization techniques. Initially introduced in 2014 as an integral component of Microsoft's Office365 service, Power BI later evolved into an independent platform and has subsequently gained immense popularity, emerging as one of the most favored BI systems available today (Power BI, n.d.). This empowers SC professionals to make data-driven decisions, optimize processes, enhance collaboration, and ultimately improve SC performance, leading to increased efficiency, reduced costs, improved customer satisfaction, and a competitive advantage in the market (Becker & Gould, 2019; Kuzmina et al., 2022). Tableau Software, SAP SE, and Qlik, among others, offer comparable services. In this research, we explored the potential of Power BI as a tool for measuring and controlling SC performance by designing a real-time dashboard that can help manage SC more effectively and affordably.

The findings of Ly Foods Ltd. (a pseudonym) presented in this research offer substantial contributions to the field of SCM by generating practical implications. These implications are derived from the study's outcomes and hold relevance for practitioners seeking to enhance their SCM practices. Due to restrictions on sharing recent data, the management of Ly Foods shared with us the 2003–2005 data. The data were mostly fine to design a dashboard using Microsoft Power BI and find management ways using its SC KPIs. This dashboard has important implications for businesses of all sizes since this research found an opportunity to manage the SC performance of a business using the ADR method. This article explores the advantages of real-time dashboards by examining the case study of Ly Foods Ltd. It highlights how this tool enhances business efficiency, visibility, accessibility, and information sharing. The ultimate outcome is improved decision-making and enhanced SC performance. To meet the aim of the research work, we have set two objectives:

- To design a simple, open, real-time, condensed, and collaborative SCM dashboard using Microsoft Power BI, and
- To show how the performance of an SC can be managed by adopting the ADR method.

The rest of the paper is structured as follows: Section 2 documents related works and motivation; Section 3 illustrates research methods; Section 4 analyzes a case of Ly Foods Ltd. Section 5 reports results and outcomes with essential discussion and key findings. Section 6 concludes the study with future research scopes, theoretical implications, and practical implications.

## 2. Related works, and motivation

### 2.1. Related works

#### 2.1.1. BI and other analytical solutions across diverse domains

To support SC analysts in making better decisions, BI includes a wide range of analytical software and solutions for acquiring, combining, analyzing, and providing access to information (Olszak, 2016). Analysts may examine trends and specifics, swiftly change inventory and distribution, spot issues with vendors' performance, and comprehend underlying SC costs and inefficiencies. Together, these activities can improve distribution and inventory techniques, SC efficiency, vendor information, and communication (Richards et al., 2019). Anderson et al. (2014) used the Statistical Analysis System (SAS) for strategic planning, marketing, and SCM because this is the most widely used advanced predictive analytics BI tool. Vuong (2020) implemented operational logistics into the company's daily business management activities and used Qlik Sense to monitor, lead, and regulate their performance. Zoho Analytics is an excellent BI tool for in-depth data and reporting analysis that can sync data periodically (Analytics, n.d.). The financial impact of the severe acute respiratory syndrome-2 disease (COVID-19) on the government because of the cancellation or postponement of numerous public projects was assessed by Ogunnusi et al. (2021) using Zoho Analytics. Marques et al. (2020) created a Microsoft Power BI-based system from a collection of SC indicators that provided better information transmission and management, which in turn improved the decision-making processes. Power BI solutions are being used by many major corporations, like Heathrow, Hewlett-Packard, Meijer, Aston Martin, Rolls-Royce, and others, to handle a variety of analytical jobs and make important business growth decisions (Iliashenko et al., 2019). Besides, BI provides a collection of predefined reports and data models for real-time reporting but no functionality for stand-alone applications (Hahn & Packowski, 2015). Models that represent tables and columns based on reports and queries were designed by Katircioglu et al. (2014) using IBM Cognos BI to optimize end-to-end pharmaceutical SC standards. Brandão et al. (2016) used the top six solutions (QlikView, Jaspersoft, Tableau, Pentaho, Spago, and Palo) in healthcare organizations to provide information to assist decision-making processes with certain applications. A benchmarking of these tools found Spago BI was the most complete open-source application compared to the software's most enterprise-focused version; it had more features. BI consists of a dynamic and ongoing collection of practices and procedures that are integrated into groups, organizations, and individuals. Oey et al. (2021) conducted a case study on a medium-sized metal manufacturer, implementing a performance measurement system using Power BI to transform Excel worksheets into visually appealing dashboards. Crestan (2022) introduced a groundbreaking decision-making dashboard solution utilizing Power BI, while Pukala et al. (2020) employed Power BI to monitor insurance activities. Libby et al. (2022) utilized Microsoft Power Pivot and Power BI for data analysis in revenue generation assessment. Kuzmina et al. (2022) employed BI solutions to oversee and enhance project activities.

#### 2.1.2. ADR method employed across different domains

The ADR methodology is encompassing, involving the development and evaluation of interventions in real-world contexts. It bridges the theory-practice gap by engaging practitioners and researchers and facilitating organizational improvement and knowledge creation. Gill and Chew (2019) shared insights from their completed ADR project in an Australian financial services organization, providing guidance for academia-industry collaborative research and contributing to the design of the configuration information system (CIS) reference architecture. Engelking et al. (2020) introduced a framework of design principles for applying machine learning (ML) to supply chain risk management (SCRM) through an ADR project conducted in partnership with an automotive company's SCRM department. Sherer (2014) emphasized the adaptation of existing theory to the unique characteristics of the healthcare industry in ADR, highlighting the generation of grounded theory through active participation in designing and evaluating socio-technical systems. Lustenberger et al. (2020) applied the ADR approach to enhance the understanding of blockchain's value in SCM by developing and evaluating artifacts in collaboration with companies and



**Table 1. A comparison of related research**

| Researchers                           | Research domain                 | Analytical solutions/tools      | Methodology                               | The outcomes                           |
|---------------------------------------|---------------------------------|---------------------------------|-------------------------------------------|----------------------------------------|
| Vuong (2020)                          | Operative logistics             | Qlik Sense                      | Analytical                                | Managing KPIs in material handling     |
| Marques et al. (2020)                 | Industry 4.0 environment        | Microsoft Power BI              | Analytical                                | Improving the decision-making process  |
| Katircioglu et al. (2014)             | SC policies                     | SC scenario modeler             | Operations research models and analytical | SC optimization                        |
| Brandão et al. (2016)                 | Healthcare environments         | Various BI tools                | Analytical                                | Finding the better BI tool             |
| Pukala et al. (2020)                  | Insurance activity              | Microsoft Power BI              | Analytical                                | Monitoring insurance activities        |
| Kuzmina et al. (2022)                 | Company's project activities    | BI solutions                    | Analytical                                | Project's quality management           |
| Tim et al. (2023)                     | Information systems (IS)        | ADAPT framework                 | ADR                                       | Support design teams with IS solutions |
| Cronholm and Göbel (2022)             | Research projects               | ×                               | ADR                                       | Managing project tasks                 |
| Clauss and Schumann (2020)            | Supply chains                   | ×                               | ADR                                       | SC collaboration                       |
| Gill and Chew (2019)                  | Financial services organization | CiS reference architecture      | ADR                                       | Enabling service resilience            |
| Engelking et al. (2020)               | Supply chains                   | Machine learning                | ADR                                       | Minimizing SC risks                    |
| Lustenberger et al. (2020)            | Supply chains                   | Blockchain technology           | ADR                                       | Adding values in SC                    |
| Herath Pathirannehelage et al. (2023) | Supply chains                   | Artificial intelligence systems | ADR                                       | SC predictive maintenance              |
| This work                             | Supply chains                   | Microsoft Power BI              | ADR                                       | Managing SC performance                |

addressing information flow issues. Cronholm and Göbel (2022) provided methodological guidance for constructing IT artifacts using the ADR method, offering procedural support, guiding concepts, and diverse techniques for documenting project tasks. Clauss and Schumann (2020) focused on developing a universal process model for implementing digital SC collaboration concepts. Herath Pathirannehelage et al. (2023) utilized ADR to create a predictive maintenance system for SC monitoring devices, enhancing decision-making regarding their maintenance. Zaitsev and Mankinen (2022) designed ADR-based financial education applications tailored for deployment in rural areas of Cambodia. Tim et al. (2023) used the ADR method and introduced ADAPT (Agility, Designation, Alignment, Participation, and Trust) framework to explore the insights of crisis-driven information systems solutions to build digital resilience. Wing et al. (2017) employed the ADR approach to guide the development, application, and evaluation of a framework. Lastly, Pan et al. (2021) utilized ADR to design sustainability principles for a wildlife management analytics system. The summarized features of the mentioned articles are presented in Table 1, which helps define the research scope of this article.

## 2.2. Motivation

The measurement of SC performance is a determinant for the use of resources and for tracking and tracing the level of activity of the SC. Besides, this measurement plays an important role in setting business goals, evaluating achievement, and determining future courses of action. Since SC

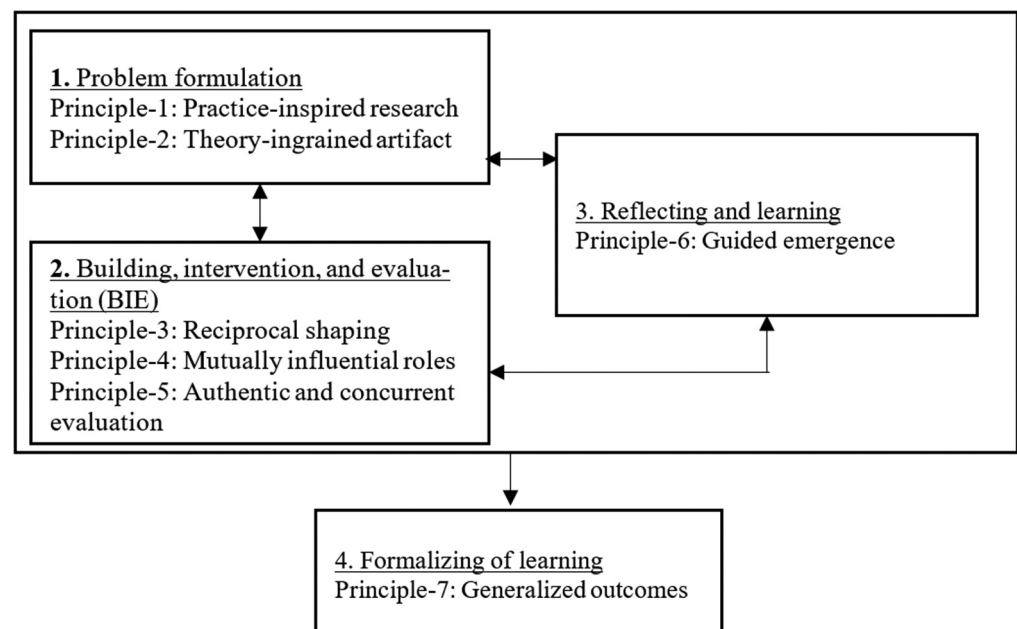
performance measurement deals with the associated strategies and technologies for managing it effectively, it has received much attention from researchers and practitioners. Previous research articles addressed various methods to measure SC performance, such as Bhattacharya et al. (2014) used fuzzy ANP-based balanced scorecard approach; Sellitto et al. (2015) used a SCOR-based model; and Qorri et al. (2018) proposed a conceptual framework to measure SC performance. Recently, the usage of dashboards developed by Microsoft Power BI is increasing in diverse fields, such as Becker and Gould (2019) used it as an extending tool of Microsoft Excel to manipulate, analyze, and visualize data; Kuzmina et al. (2022) found BI solutions as a tool for quality management in a company; Powell (2018), Widjaja and Mauritsius (2019) found this as a BI tool as well as a visualization tool, respectively. The usage of Microsoft Power BI is gaining significant momentum across diverse sectors, including research projects, service organizations, healthcare service centers, logistics, and others. However, no substantial research has solely focused on managing SC performance through the ADR method utilizing the dashboard provided by Microsoft Power BI. Hence, this has motivated the authors to conduct this research, particularly on this topic.

### 3. Methodology

#### 3.1. ADR method for the scenario under study

The methodology section was carried out using Sein et al. (2011)'s ADR method, as shown in Figure 1. This research focused on the difficulties associated with lower SC performance, such as difficulties in inventory control, demand forecasting, customer demand prediction, service quality, etc. There might be several causes of lower SC performance, and it is important to see how these causes affect it. Consequently, this article moves forward by utilizing data from Ly Foods Ltd., a prominent food manufacturing company in Bangladesh, to visualize its impact from a management perspective. The objective is to identify the underlying reasons behind poor performance and guide the management in implementing effective solutions. However, the complexities involved, including the types of challenges, their characteristics, and the appropriate preventive or corrective actions, align well with the ADR method. Therefore, this research employs the following iterative cycles: problem formulation; building, intervention, and evaluation (BIE); reflecting and learning; formalizing learning.

Figure 1. Stages and principles of ADR method.



### 3.2. Problem formulation: difficulties in measuring SC performance as a class of problems

Since the article aimed to measure the SC performance of Ly Foods Ltd. and provided ways or guidelines to control its performance, we considered some difficulties in measuring SC performance as a class of problems. This step of problem formulation is the first step of the ADR method and suggests the following tasks to be done (Sein et al., 2011). Firstly, the identification and conceptualization of research opportunities such as customer dissatisfaction, improper resource utilization, lower efficiency, and wrong forecasting are shown in Table 2. Secondly, the formulation of research problems and limitations. Thirdly, casting the problem as an instance of a class of problems that is as similar as the scenario of the fitness fall of a manufacturing unit as described by Rahman and Rahman (2019, 2020). Fourthly, identification of theoretical bases that are as similar as the theoretical bases of Rahman and Rahman (2019) and uses of prior tools and technologies, such as QlikView, Jaspersoft, Spago, Palo, etc. Besides, this article secures long-term organizational commitments and sets up roles and responsibilities for the SCM based on the learnings of Ly Foods Ltd.

### 3.3. BIE: Microsoft Power BI at Ly Foods' SC

Since decisions about designing, shaping, and reshaping the ensemble SC performance factors and intervening in organizational work practices should be interwoven with ongoing evaluation, the BIE follows the principles of step 2 of Figure 1. Microsoft Power BI as a dominant component of Ly Foods' SC and its detailed view have been shown in Figures 2 and 3, respectively. However, to ensemble these factors with organizational practices, this article proposes a five-layered BI architecture to design a dashboard. This dashboard will be used to measure and control SC performance using the data from SC performance factors. Now, the layers are: (1) the data source layer; (2) the Extract-Transform-Load (ETL); (3) the data warehouse; (4) the metadata layer; and (5) the end-user layer. However, to assess this tool (Microsoft Power BI) and to analyze Ly Foods' SC performance, this research executed the BIE cycle in sections 3.4 and 3.5.

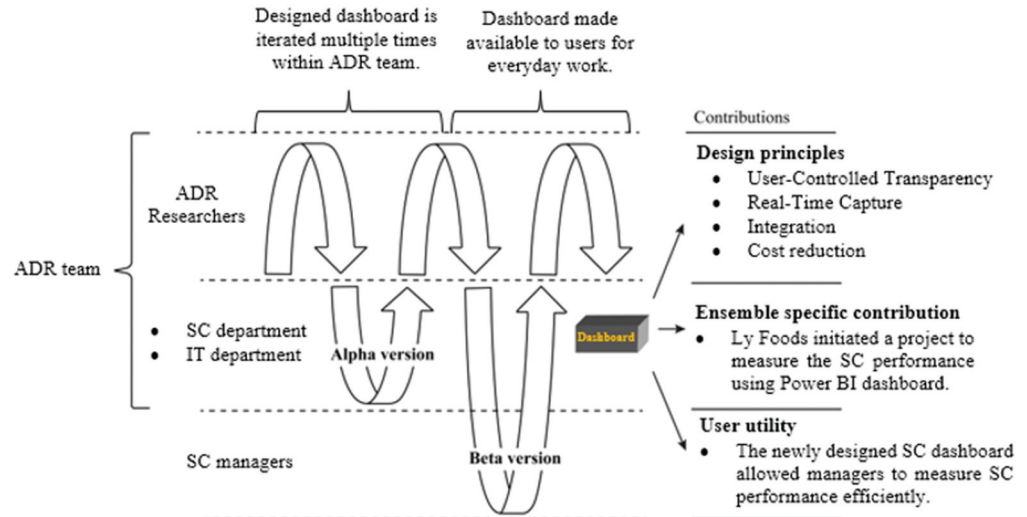
#### 3.3.1. Data source layer

The layers are shown in Figure 4, and they are ETL (Extract-Transform-Load), data source, data warehouse, end user, and metadata layers. Information that is gathered and kept up-to-date using various operational systems, such as CRM and ERP, is referred to as an internal data source. Data on customers, products, and sales is one of the sources of internal data (Baars & Kemper,

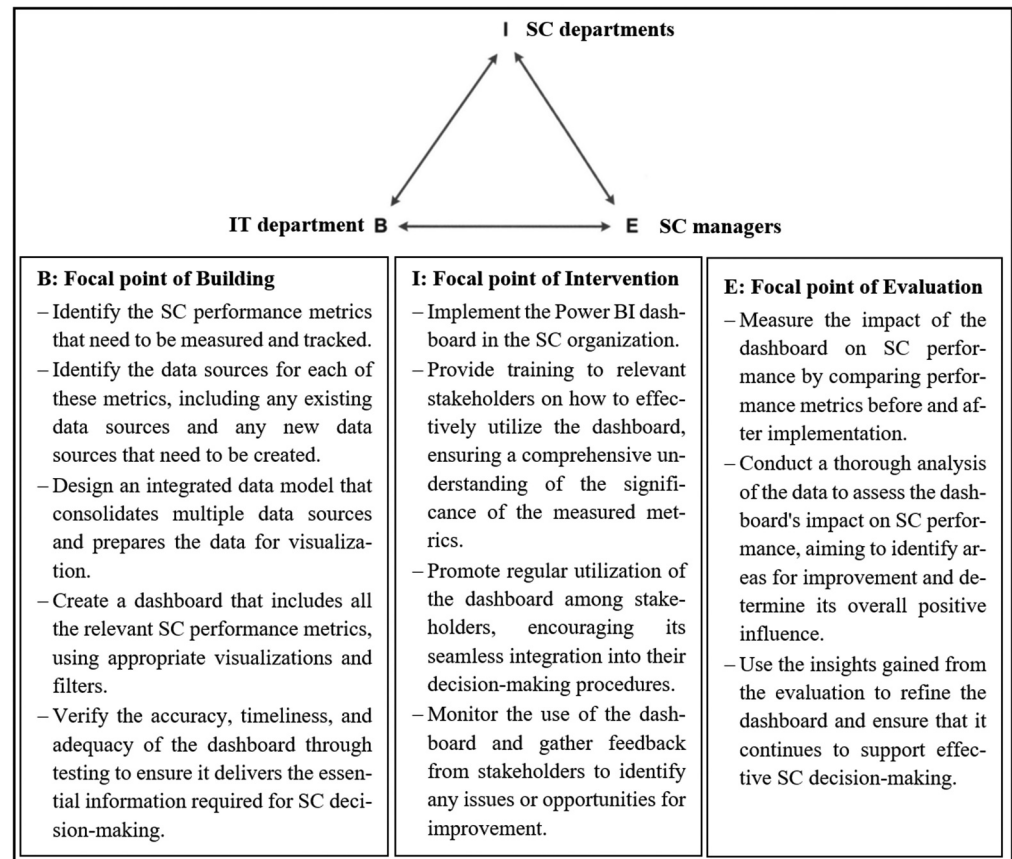
**Table 2. Difficulties in measuring SC performance and current practices**

| Difficulties in measuring SC performance                                                                                                                                                            | Current practices                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>• Forecasting</li> <li>• Resources utilization</li> <li>• Information sharing</li> <li>• Inventory controlling</li> <li>• Meeting customer demand</li> </ul> | <p>Nowadays, various software, tools, and dashboards are used to monitor SC performance. But there are some issues, such as cost, flexibility, and user interface, that must be taken into consideration because they influence SC performance. For example, Oracle demand planning struggles to adjust changes in demand due to its reliance on historical data and limited ability to incorporate real-time data (Edwards, n.d.); SAP SCM requires specialized training to use effectively, which limits its ability to provide real-time visibility into resource utilization (Knolmayer et al., 2009); and IBM Sterling SC Business Network lacks data security, resulting in unauthorized access and data breaches. Manhattan Associates' Warehouse Management System faces challenges in real-time inventory allocation optimization due to its reliance on pre-defined rules and processes (Wang et al., 2009). Blue Yonder faces difficulties in real-time demand adjustment, resulting in stockouts or overstocking (Alicke et al., 2017). Moreover, both software and applications have limitations and are expensive to implement.</p> |

**Figure 2. SC-dominant BIE in the Ly Foods' SC.**



**Figure 3. Detailed view of BIE at Ly Foods Ltd.**

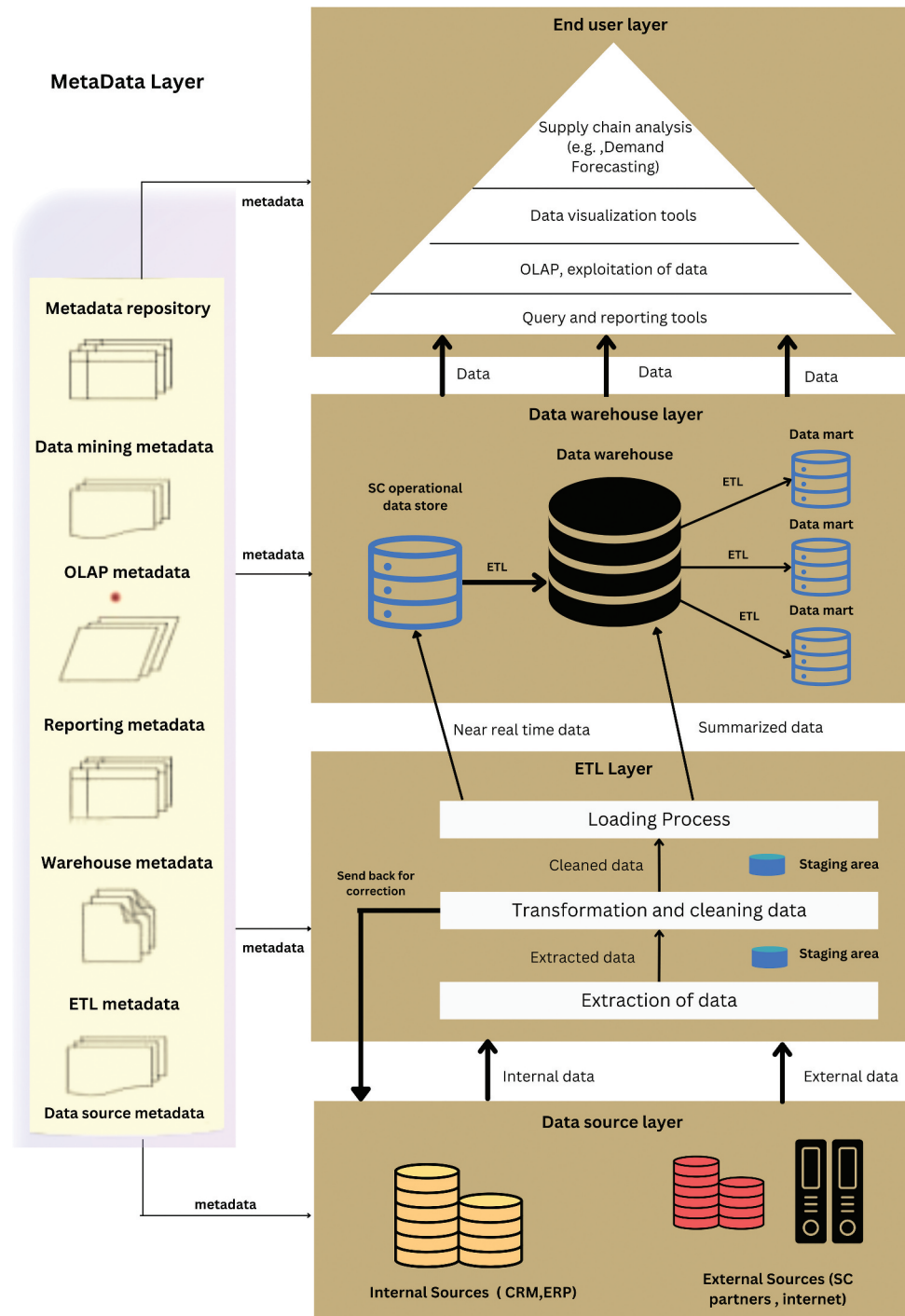


2008). Organizations are required to clearly state where their data comes from. Knowing where to get data enables a quicker and more efficient resolution of business issues.

### 3.3.2. ETL layer

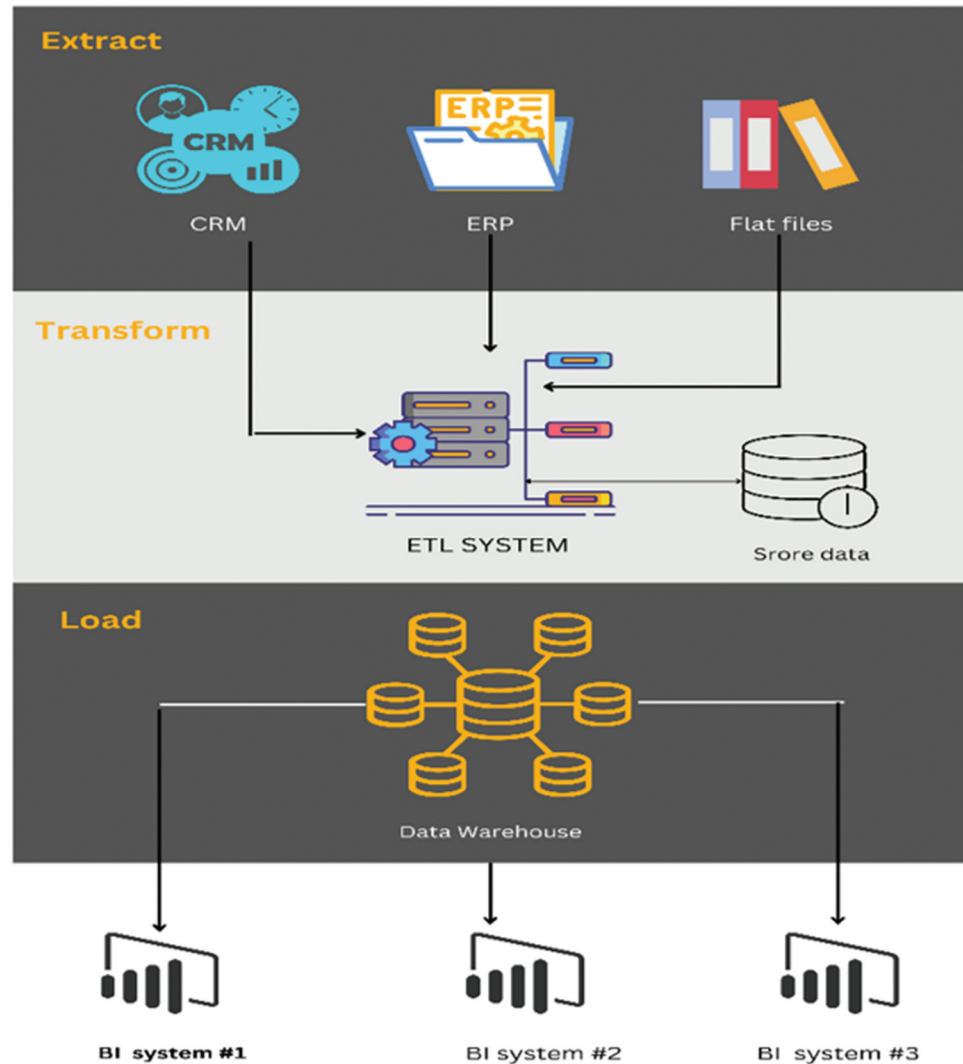
The extraction of data from different sources is known as extraction. Sometimes internal and external data are not integrated; sometimes those are incomplete or duplicated. Thus, data

**Figure 4. BI framework.**



extraction is required to pick relevant data for organizational decision-making. To transform and clean up the data, the retrieved files are then transmitted to a data staging point. If a problem emerges, this prevents the need to extract data again. Finding and fixing data errors in accordance with predetermined rules is the process of data cleansing. Errors in extracted data are returned to the original data source for correction if they are present. Following transformation and purification, data is staged. If the loading process fails, there is no need to change the data again. The

**Figure 5. ETL conceptual process.**



loading phase completes the ETL procedure. In the target repository, they are loaded. The whole process of ETL functioning is shown in Figure 5.

### 3.3.3. Data warehouse

An internal and external data warehouse gathers and saves information for strategic decision-making, querying, and analysis. A data warehouse stores aggregated data. For lengthy analysis, it also keeps a ton of past data. Weekly or daily updates are made based on the data in a data warehouse.

### 3.3.4. Metadata layer

Metadata provides descriptions of data, including their relationships, sources, usage, storage, and any modifications made. A metadata repository houses data definitions and business rules. Development time can be decreased by effective metadata management. Because they are saved as metadata, users do not need to redesign data structures for data modeling, such as table names and data types. Users can access the information by querying repositories. As a result, metadata in repositories must be updated frequently. Metadata facilitates the extraction, transformation, and loading of data.



### 3.3.5. End user layer

In the end-user layer, there are tools that allow users to see data in various formats. A pyramidal arrangement of these tools is possible. As you progress up the pyramid from the bottom, data processing and display become more complex. Making decisions gets harder when organizational structures get more complex. Operational management, for instance, uses query and reporting tools at the base of the pyramid, whereas top management utilizes analytical applications at the pyramid's apex.

### 3.4. Reflecting and learning

The third step is to implement the solutions, which require describing KPIs, connecting the data sources, and using the Microsoft Power BI dashboard. This research describes the implementation process, including data preparation, dashboard development, and deployment. This process follows the following steps: defining KPIs/managing factors; connecting data sources to Power BI; transforming and shaping data; and designing the dashboard.

#### 3.4.1. Defining KPIs/managing factors

Identifying the KPIs of an SC is very important to implementing the solutions adopted by the ADR method. These KPIs include various metrics such as inventory turnover, order cycle time, delivery performance, etc. that play significant roles in managing SC performance. Now, this article lists the KPIs for Ly Foods Ltd. that are shown in Table 3.

#### 3.4.2. Connect data sources to Power BI

After identifying the KPIs for Ly Foods SC, the subsequent step involves connecting data sources to Power BI to assess the corresponding performance. This connectivity encompasses various sources such as Excel spreadsheets, SQL Server databases, SharePoint lists, and more.

#### 3.4.3. Transform and shape data

As Power BI provides an intuitive interface to transform and shape data through a variety of options, we eliminated any inconsistencies or redundancies in the data and transformed it into a more manageable format, allowing for more effective analysis. Additionally, Power BI's data transformation features help users identify patterns, trends, and outliers that can provide valuable insights into SC performance.

#### 3.4.4. Design the dashboard

Including the relevant KPIs such as inventory turnover, order cycle time, delivery performance, and other parameters, we designed a dashboard as shown in Section 3.5. It was user-friendly and allowed quick access to critical information. Besides, Power BI's drag-and-drop interface simplifies this process by allowing users to customize the dashboard's layout and style according to their

**Table 3. KPIs and their roles in managing the performance of Ly Foods' SC**

| KPIs/managing factors        | Actions to manage SC performance                                                                              |
|------------------------------|---------------------------------------------------------------------------------------------------------------|
| Inventory turnover           | Act if inventory replenishment is not completed within the specified lead time.                               |
| Order cycle time             | Initiate replenishment measures if inventory restocking is not completed within the specified lead time.      |
| Order status                 | Implement the necessary measures to address delayed shipments and cancelled orders.                           |
| Order quantity               | Take the necessary actions whenever the order quantity significantly deviates from the specified requirement. |
| Rank of the product by sales | Initiate actions for the top-performing or underperforming product items in terms of sales.                   |
| Sales by product             |                                                                                                               |
| Sales by country             |                                                                                                               |

preferences. Now, this is providing real-time insights into SC performance, helping business owners make informed decisions based on accurate and up-to-date information.

### **3.5. Formalizing of learning**

The final step is to evaluate the effectiveness of the possible solutions and measure their impacts on the real-time dashboard. This research shows how the KPIs mentioned in Table 3 can be used to measure SC performance. The dashboard is thoroughly analyzed to interpret its effectiveness with a case study of Ly Foods Ltd.

#### **3.5.1. Analyze dashboard data to measure SC**

Once the dashboard has been designed in Power BI, the data can be analyzed to gain insights into SC performance. With the help of various visualizations, such as bar charts, line graphs, and pie charts, the dashboard will provide a clear picture of the performance of different SC metrics, such as inventory turnover, order cycle time, and delivery performance. By analyzing data, business owners can identify areas where they need to improve their SC processes and make data-driven decisions to optimize SC performance.

#### **3.5.2. Interpretation**

The data that was analyzed in the previous sections will be interpreted to draw conclusions about the effectiveness of the real-time dashboard to improve SC performance. The results will be synthesized, considering broader implications.

#### **3.5.3. Feedback and adjustment**

Based on the results from the evaluation section, if needed, the real-time dashboard will be adjusted. Feedback from users and stakeholders will be considered to identify areas for improvement and guidelines for future development.

The ADR method provides a framework for conducting research that combines problem-solving with theory development. By following the steps shown in sections 3.2 to 3.5, this article provided a comprehensive analysis of how a real-time Microsoft Power BI dashboard can improve SC performance through a case study.

## **4. A case study**

### **4.1. SCM dashboard**

After completing the above-mentioned steps, this article shows an output page for Ly Foods in Figure 6. This is the main landing page for the entire SC. This contains three main pages, i.e., an overview, order details, and product analysis. The following sections will dive deep into each of them.

### **4.2. Interpretation of the dashboard**

#### **4.2.1. Overview page**

The overview page of Ly Foods displays key information such as the number of orders, sales quantity, order numbers, and orders from different countries between two given dates. The top row of the overview page is divided into six sections, which are shown in Figure 7. Each provides specific information to aid decision-making. The first section displays the total number of orders received by the seller. The second section shows the sum of sales generated from fulfilling those orders. The third section indicates the total quantity of products included in the orders. The fourth section reveals the count of unique products ordered. The fifth section highlights the number of countries from which the seller receives orders. Additionally, there is a slicer section that allows users to select specific dates or date ranges to view precise data for that period.

The middle row is divided into three portions: (1) quantity ordered by product, which stacks on a column chart; (2) count of order number by status, which shows the status of the order; and (3)

Figure 6. SCM dashboard.

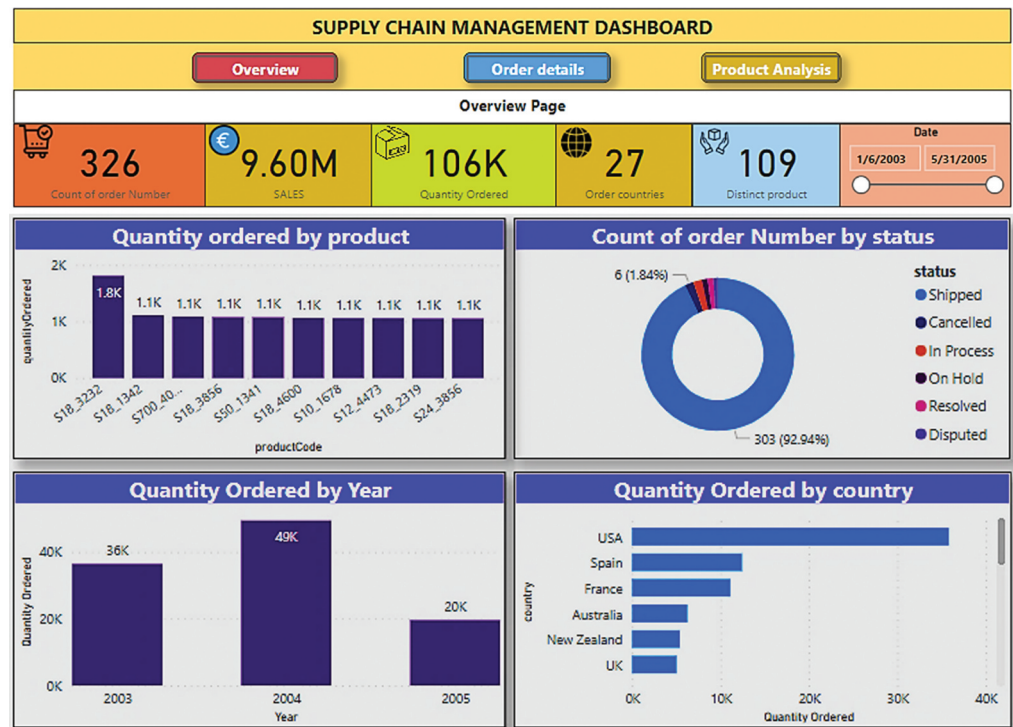


Figure 7. Top row of overview page.

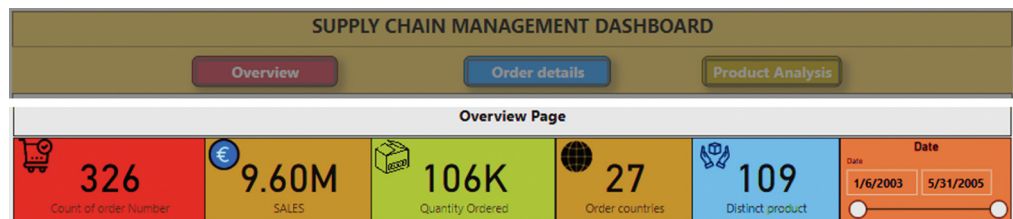
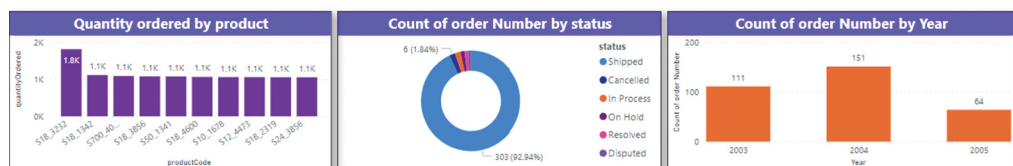


Figure 8. Middle row of overview page.



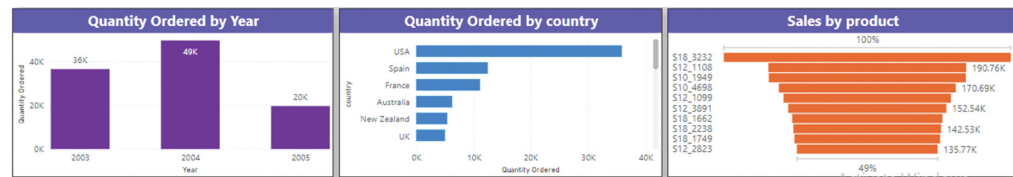
count of order number by year, which shows the total number of orders dropped. The division of the middle row and its function have been shown in Figure 8.

Similarly, the bottom row is divided into three portions: (1) quantity ordered by year, which shows the total quantity of different products ordered in a specific year; (2) quantity ordered by country, which shows the size of order quantities from different countries; and (3) sales by product, which shows the sales generated by each product-by-product code. The division of the bottom row and its function have been shown in Figure 9.

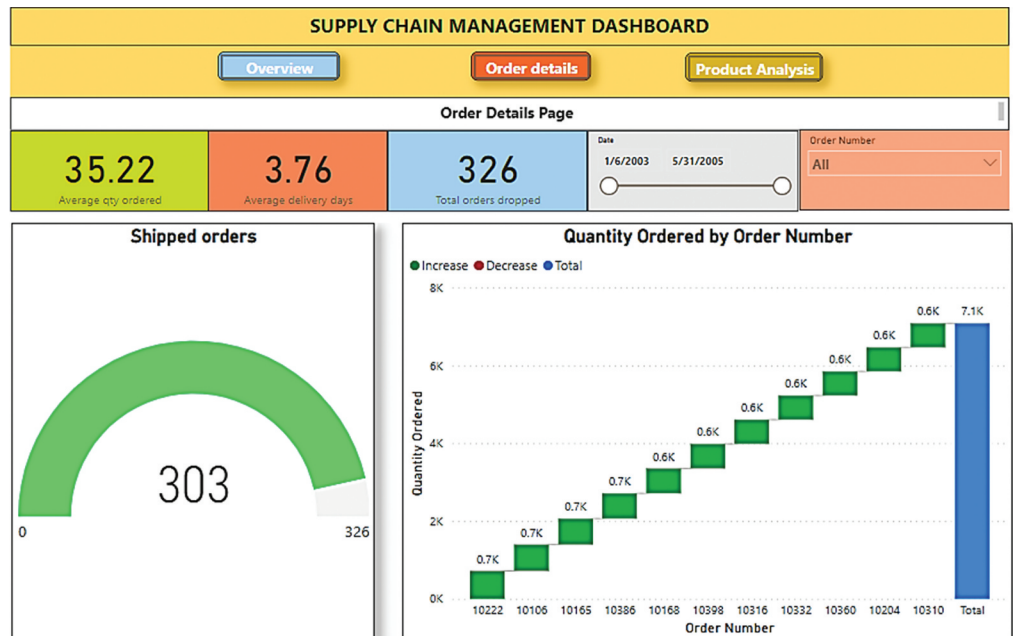
#### 4.2.2. Order details page

The order details page provides all the essential information for Ly Foods SC, such as delivery date, order drop date, number of shipped products, and sales, as shown in Figure 10. This page is

**Figure 9. Bottom row of over-view page.**



**Figure 10. Order details page.**



important for Ly Foods in managing its SC as it provides an in-depth view of each order, allowing businesses to monitor and manage their SC operations more efficiently.

The top row of the order details page of Figure 11 has five portions: (1) average order quantity, displaying the average size of each order; (2) average delivery date, indicating the seller's lead time; (3) total orders placed with the seller; (4) order number selection for accessing order details; and (5) date slicer, for selecting specific dates or date ranges to view precise data.

The bottom row of Figure 12 consists of three portions: (1) shipped orders, indicating the number of orders shipped compared to total orders; (2) quantity ordered, displaying the total quantity of orders based on the order number; and (3) sales by order number, representing the sales generated from different orders.

#### 4.2.3. Product analysis page

The product analysis page is also important because it provides business insight into the performance of their products, as shown in Figure 13. This page displays comparisons between products,

**Figure 11. Top row of order details page.**

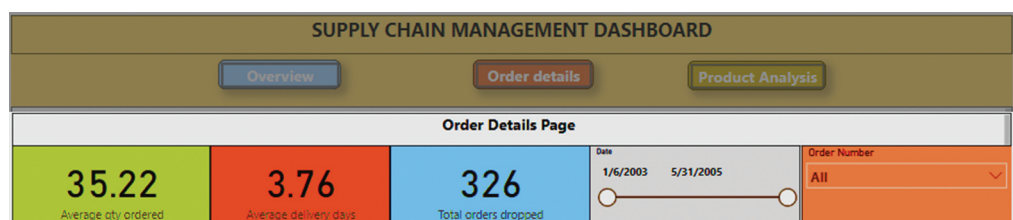
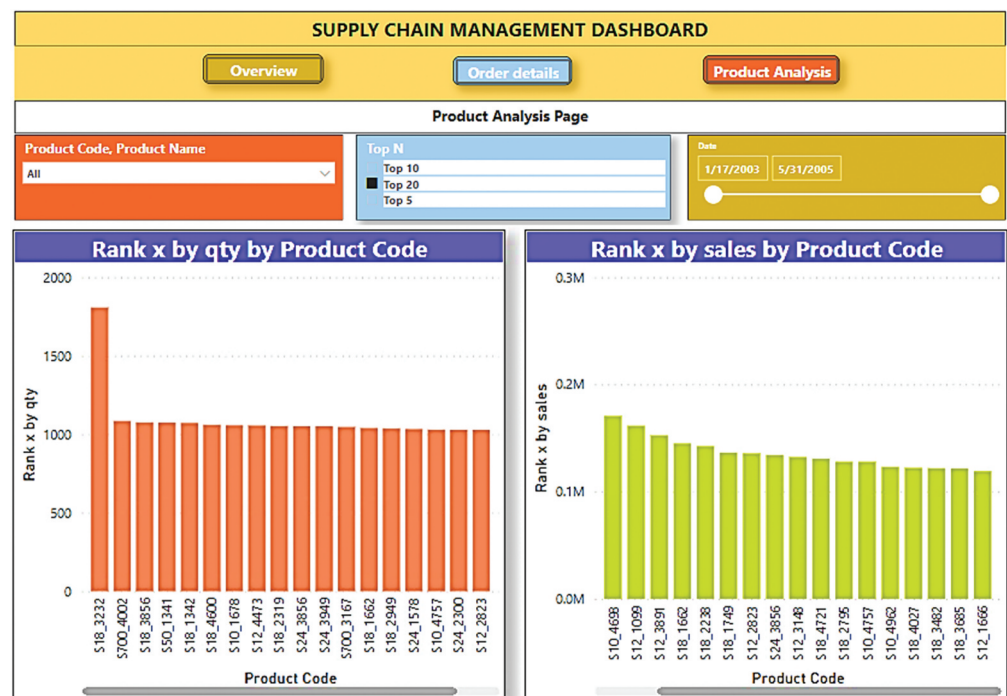


Figure 12. Bottom row of order details page.



Figure 13. Product analysis page.

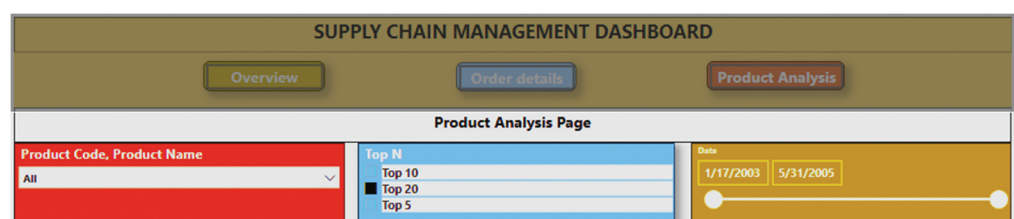


including their sales or quantity, and ranks them accordingly. By analyzing the performance of the products of Ly Foods, suppliers make data-driven decisions to optimize their SC operations.

The top row of Figure 14 has three sections: (1) product code and name for selecting specific product information; (2) top N for specifying the display of the top 10, 20, or 5 products on the dashboard; and (3) date slicer for selecting specific dates or date ranges to view precise data.

The bottom row of Figure 15 has three parts: (1) rank x by quantity by product code, ranking products based on order quantity by product code; (2) rank x by sales by product code, ranking

Figure 14. Top row of product analysis page.





**Figure 15. Bottom row of product analysis page.**



products based on sales generated by each product-by-product code; and (3) sum price by quantity order, illustrating the relationship between order size and total price.

## 5. Results and discussion

This study conducted a case study on Ly Foods' SC to manage its performance using a Microsoft Power BI dashboard. The dashboard features three main pages, namely the overview, order details, and product analysis pages. The overview page offers a comprehensive summary of the entire system, allowing users to track real-time updates on KPIs such as sales, orders, and orders from various countries. For example, from the Ly Foods' dashboard overview page, users can see that Ly Foods generated \$9.60 million in sales from 326 orders, providing them with a means of evaluating whether they have met their sales target. The order details page provides users with a detailed view of all orders, including delivery dates, order drop dates, shipped products, and sales generated from each order. For instance, users can view the average delivery time for a product, which is 3.76 days, providing valuable insights into Ly Foods' lead time. The product analysis page compares and ranks products based on their sales or quantity. For example, if readers select the "S10\_1678" product code, they can easily track that a total of 1,000 pieces of this product were sold, generating almost \$0.1 million in sales. By analyzing the performance of their products and monitoring and managing the details of each order, businesses can identify areas that need improvement, optimize their operations, and improve their overall performance. The dashboard provides businesses with real-time insights into their SC operations, enabling them to make data-driven decisions that optimize their operations. A summary of the performance of Ly Foods SC is shown in Table 4, where the ADR process was applied.

By leveraging the clean data storage capabilities and expanding upon Power BI's core graphical and statistical features, suppliers are empowered to gain comprehensive insights through a unified report. This approach finds support in the works of Powell (2017, 2018). With access to diverse tabular data in various formats, ranging from extensive databases to simple text files, suppliers can employ Power Query, a foundational layer of Power BI, to reformat and combine the data prior to loading it into Excel. This grants them enhanced control over data types and delimiters and enables sophisticated transformations, surpassing the limitations of traditional text import features. According to Becker and Gould (2019), Power BI extends three key advantages to suppliers: (1) the availability of standalone Windows applications with visualization and analysis capabilities; (2) a cloud platform for sharing within the organization, with or without desktop software; and (3) reduced per-user costs. Moreover, it facilitates data retrieval, cleaning, and shaping processes, enabling suppliers to focus specifically on the data required for effective performance management within a particular project.

## 6. Conclusion, implications, and future directions

To be more competitive and to implement strategies based on SC KPIs, a real-time Microsoft Power BI dashboard plays an important role. Hence, this research has presented a practical solution for small businesses to improve their measurement through the development of a real-time dashboard, where a case study on Ly Foods Ltd. was conducted. This study has also demonstrated the potential benefits of using innovative technologies in SCM practices, providing more efficient, visible, accessible, and



**Table 4. Summary of the ADR process in the Ly Foods' SC performance measurement**

| Stages and Principles                            |                                                                                                                                                                                                                                             | Dashboard                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|--------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Stage 1: Problem Formulation                     |                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Principle 1: Practice-Inspired Research          | Research was driven by the need for better measurement of SC performance in a production-based organization.                                                                                                                                | <b>Recognition:</b> A real-world problem or challenge related to the measurement of SC performance that needed to be addressed was recognized as lacking.                                                                                                                                                                                                                                                                                                                        |
| Principle 2: Theory-Ingrained dashboard          | The dashboard was designed to reflect the relevant theories and concepts that inform the measurement and management of SC performance.                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Stage 2: BIE                                     |                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Principle 3: Reciprocal Shaping                  | SC managers and practitioners provided feedback and input throughout the BIE stage to ensure that the dashboard was tailored to their needs and interests.                                                                                  | <b>Alpha Version:</b> The alpha version of the dashboard was the first version developed in the ADR process. This version was designed to be a basic prototype that would allow the researcher to test and refine the dashboard's functionality, usability, and design in collaboration with practitioners in the field.<br><b>Beta Version:</b> The beta version was designed to be tested in a real-world setting, with practitioners using the dashboard in their daily work. |
| Principle 4: Mutually Influential Roles          | The researcher and practitioners worked together to co-create the dashboard and evaluate its effectiveness.                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Principle 5: Authentic and Concurrent Evaluation | The evaluation was based on authentic performance data and involved ongoing feedback and iteration in the context of the SC performance measurement.                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Stage 3: Reflection and Learning                 |                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Principle 6: Guided Emergence                    | Insights gained from the BIE stage are used to refine and develop a new dashboard for the measurement and management of SC performance.                                                                                                     | <b>Emerging version and realization:</b> The emerging version, the third version, was tested by a group of practitioners in a real-world setting, with feedback collected through surveys and interviews. Feedback from this testing was used to further refine the dashboard and develop the realization version.                                                                                                                                                               |
| Stage 4: Formalization of Learning               |                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Principle 7: Generalized Outcomes                | The knowledge and insights gained from the ADR process can be generalized to other contexts and settings. This study demonstrated how the theory and insights gained from the ADR process can be applied to other SC contexts and settings. | <b>Ensemble version:</b> This version was the culmination of the previous versions, incorporating all the feedback and refinements made throughout the ADR process. The ensemble version was designed to be a fully functional and effective tool for measuring and managing SC performance.                                                                                                                                                                                     |

shareable information for better decision-making and improved SC performance. By presenting a practical solution for small businesses to improve their SC performance measurement, this research offers valuable contributions to the field of SCM. The designed dashboard provides simple, open, real-time, condensed, and collaborative SC solutions. Besides, this shows ways to manage SC performance by taking corrective or preventive actions for a KPI (see Tables 2 and 3). Moreover, it helps manage an SC more effectively and affordably, leading to improved operational efficiency, customer satisfaction, and profitability. However, this article specifies the theoretical and practical implications of this research, which are more precisely described below. Finally, this article paves the way for further studies on using more advanced BI tools for SC performance measurement and using multiple data sources to create more comprehensive dashboards that enable businesses to better compete in the global market.

Regarding further research scope, this paper provides a foundation for future studies on using more advanced BI tools for SC performance measurement. For instance, future research can explore the use of machine learning algorithms to analyze data from multiple sources and create more comprehensive dashboards that enable businesses to better compete in the global market. Additionally, future studies can examine the adoption and implementation of real-time SC performance measurement dashboards across various industries and regions.

### 6.1. Theoretical contribution

This research makes a significant theoretical contribution by showcasing the potential advantages that arise from incorporating innovative technologies into SC practices. The study emphasizes the significance of measuring SC performance and explores a cost-effective and easily accessible solution tailored for small businesses. The development of a real-time dashboard offers a pragmatic approach to measuring SC KPIs, which can prove invaluable to small businesses lacking the means to invest in expensive SCM systems.

### 6.2. Practical implications

The practical implications of this research are also noteworthy. This dashboard finds out which KPIs are showing lower performance or are responsible for the overall lower performance of an SC. Applying the ADR method in a similar way as this article applied to the case of Ly Foods, the suppliers can design SC artifacts adopting a new SC-dominant BIE to manage performance. Additionally, there is an opportunity to assess the performance of a BIE at the alpha level before advancing to the beta level. Based on this performance evaluation, individuals can make an informed decision on whether to proceed or not. Finally, by enabling businesses to make data-driven decisions and monitor their SC performance regularly, the dashboard facilitates better SCM practices and enhances overall performance.

#### Acknowledgment

The publication of this article was funded by the Qatar National Library.

#### Funding

The authors declare that no funds, grants, or other support were received for this research work.

#### Author details

Dewan Hafiz Nabil<sup>1</sup>  
 ORCID ID: <http://orcid.org/0000-0003-2499-5420>  
 Md. Habibur Rahman<sup>1,2</sup>  
 E-mail: [mdra47330@hbku.edu.qa](mailto:mdra47330@hbku.edu.qa)  
 ORCID ID: <http://orcid.org/0000-0003-3860-3678>  
 Altaf Hussain Chowdhury<sup>1</sup>  
 Brenno Castrillon Menezes<sup>2</sup>  
 ORCID ID: <http://orcid.org/0000-0002-2972-7014>  
<sup>1</sup> Department of Industrial Engineering and Management, Khulna University of Engineering & Technology, Khulna, Bangladesh.  
<sup>2</sup> Division of Engineering Management and Decision Sciences, Hamad Bin Khalifa University, Doha, Qatar.

#### Disclosure statement

No potential conflict of interest was reported by the author(s).

#### Data availability statement

The authors confirm that the data supporting the findings of this study are available within the article and/or its supplementary materials.

#### Correction

This article has been corrected with minor changes. These changes do not impact the academic content of the article.

#### Citation information

Cite this article as: Managing supply chain performance using a real time Microsoft Power BI dashboard by action design research (ADR) method, Dewan Hafiz Nabil, Md. Habibur Rahman, Altaf Hussain Chowdhury & Brenno Castrillon Menezes, *Cogent Engineering* (2023), 10: 2257924.

#### References

- Alicke, K., Glatzel, C., Hoberg, K., & Karlsson, P. M. (2017). *Big data and the supply chain: The big supply chain analytics landscape*. McKinsey & Company.
- Analytics. (n.d.). *Modern self-service BI and analytics platform*. <https://www.zoho.com/analytics/>
- Anderson, J. E., Daniels, J., McDonald, D., & Edvalson, R. (2014). The current state of business intelligence and analytics in Utah. *Issues in Information Systems*, 15 (2), 359–366. [https://doi.org/10.48009/2\\_iis\\_2014\\_359-366](https://doi.org/10.48009/2_iis_2014_359-366)
- Baars, H., & Kemper, H. G. (2008). Management support with structured and unstructured data—an integrated business intelligence framework. *Information Systems Management*, 25(2), 132–148. <https://doi.org/10.1080/10580530801941058>
- Becker, L. T., & Gould, E. M. (2019). Microsoft Power BI: Extending excel to manipulate, analyze, and visualize diverse data. *Serials Review*, 45(3), 184–188. <https://doi.org/10.1080/00987913.2019.1644891>
- Bhattacharya, A., Mohapatra, P., Kumar, V., Dey, P. K., Brady, M., Tiwari, M. K., & Nudurupati, S. S. (2014). Green supply chain performance measurement using fuzzy ANP-based balanced scorecard: A collaborative decision-making approach. *Production Planning & Control*, 25(8), 698–714. <https://doi.org/10.1080/09537287.2013.798088>

- Brandão, A., Pereira, E., Esteves, M., Portela, F., Santos, M. F., Abelha, A., & Machado, J. (2016). A benchmarking analysis of open-source business intelligence tools in healthcare environments. *Information*, 7(4), 57. <https://doi.org/10.3390/info7040057>
- Clauss, A. M., & Schumann, C. A. (2020). A systems theory and action design research perspective on supply chain collaboration in the context of SCM 4.0. In 2020 *IEEE International Conference on Engineering, Technology and Innovation (ICE/ITMC)* (pp. 1–8). IEEE. <https://doi.org/10.1109/ICE/ITMC49519.2020.9198596>
- Crestan, L. (2022). *Implementation of a dashboard for measuring warehouse performance-proposal based on a business intelligence software*. <http://hdl.handle.net/10589/187128>
- Cronholm, S., & Göbel, H. (2022). Action design research: Integration of method support. *International Journal of Managing Projects in Business*, 15(8), 19–47. <https://doi.org/10.1108/IJMPB-07-2021-0196>
- Edwards, O. J. (n.d.) Oracle value chain planning Demantra demand management. <https://gazelle.in/wp-content/uploads/2017/04/Demantra-Demand-Management.pdf>
- Engelking, B., Buchholz, W., & Köhne, F. (2020). Design principles for the application of machine learning in supply chain risk management: An action design research approach. *Supply Management Research: Aktuelle Forschungsergebnisse*, 2020, 137–162. [https://doi.org/10.1007/978-3-658-31898-7\\_8](https://doi.org/10.1007/978-3-658-31898-7_8)
- Gill, A. Q., & Chew, E. (2019). Configuration information system architecture: Insights from applied action design research. *Information & Management*, 56(4), 507–525. <https://doi.org/10.1016/j.im.2018.09.011>
- Hahn, G. J., & Packowski, J. (2015). A perspective on applications of in-memory analytics in supply chain management. *Decision Support Systems*, 76, 45–52. <https://doi.org/10.1016/j.dss.2015.01.003>
- Hannila, H., Kuula, S., Harkonen, J., & Haapasalo, H. (2022). Digitalisation of a company decision-making system: A concept for data-driven and fact-based product portfolio management. *Journal of Decision Systems*, 31(3), 258–279. <https://doi.org/10.1080/12460125.2020.1829386>
- Herath Pathirannehelage, S., Johannsson, J. G., Shrestha, Y. R., & von Krogh, G. (2023). Artificial intelligence-augmented decision making in supply chain monitoring: An action design research study. *European Conference on Information Systems*, (pp. 1–17). Available at SSRN: <https://ssrn.com/abstract=4422594>
- Iliashenko, O., Iliashenko, V., & Esser, M. (2019). BI systems implementation for supply chain sector in retail companies. In *International Conference on Digital Technologies in Logistics and Infrastructure (ICDTLI 2019)* (pp. 299–305). Atlantis Press. <https://doi.org/10.2991/icdtli-19.2019.53>
- Katircioglu, K., Gooby, R., Helander, M., Drissi, Y., Chowdhary, P., Johnson, M., & Yonezawa, T. (2014). Supply chain scenario modeler: A holistic executive decision support solution. *Interfaces*, 44(1), 85–104. <https://doi.org/10.1287/inte.2013.0725>
- Knolmayer, G. F., Mertens, P., Zeier, A., & Dickersbach, J. T. (2009). *Supply chain management based on SAP systems: Architecture and planning processes*. Springer Science & Business Media. <https://doi.org/10.1007/978-3-540-68739-9>
- Kuzmina, S. N., Evdokimova, A. A., & Kondakova, V. V. (2022). BI solutions as a tool for the management of quality of company's project activities. In 2022 *International Conference on Quality Management, Transport and Information Security, Information Technologies (IT&QM&IS)* (pp. 296–299). IEEE. <https://doi.org/10.1109/ITQMIS56172.2022.9976648>
- Libby, T., Schwebke, J. M., & Goldwater, P. M. (2022). Using data analytics to evaluate the drivers of revenue: An introductory case study using Microsoft Power pivot and Power BI. *Issues in Accounting Education*, 37(4), 97–105. <https://doi.org/10.2308/ISSUES-2021-057>
- Lustenberger, M., Spychiger, F., & Malesevic, S. (2020). Towards a better understanding of the value of blockchains in supply chain management. In *Information Systems: 16th European, Mediterranean, and Middle Eastern Conference, EMCIS 2019, Dubai, United Arab Emirates*, (pp. 101–112). Springer International Publishing. [https://doi.org/10.1007/978-3-030-44322-1\\_8](https://doi.org/10.1007/978-3-030-44322-1_8)
- Marques, R., Moura, A., & Teixeira, L. (2020). Decision support system for the industry 4.0 environment: Design and development of a business intelligence tool. *Proceedings of the International Conference on Industrial Engineering and Operations Management*, Detroit, Michigan, USA (pp. 1613–1624).
- Oey, E., Harno, S. S. S., & Zain, C. (2021). Developing integrated performance dashboards with Power BI—a case study in a medium-size Manufacturer. In 2021 *International Conference on Information Management and Technology (ICIMTech)*, Vol. 1, (pp. 265–270). IEEE. <https://doi.org/10.1109/ICIMTech53080.2021.9535105>
- Ogunnusi, M., Omotayo, T., Hamma-Adama, M., Awuzie, B. O., & Egbelakin, T. (2021). Lessons learned from the impact of COVID-19 on the global construction industry. *Journal of Engineering, Design & Technology*, 20(1), 299–320. <https://doi.org/10.1108/JEDT-05-2021-0286>
- Olzak, C. M. (2016). Toward better understanding and use of business intelligence in organizations. *Information Systems Management*, 33(2), 105–123. <https://doi.org/10.1080/10580530.2016.1155946>
- Pan, S. L., Li, M., Pee, L. G., & Sandeep, M. S. (2021). Sustainability design principles for a wildlife management analytics system: An action design research. *European Journal of Information Systems*, 30(4), 452–473. <https://doi.org/10.1080/0960085X.2020.1811786>
- Powell, B. (2017). *Microsoft Power BI cookbook: Creating business intelligence solutions of analytical data models, reports, and dashboards*. Packt Publishing Ltd.
- Powell, B. (2018). *Mastering Microsoft Power BI: Expert techniques for effective data analytics and business intelligence*. Packt Publishing Ltd.
- Power BI. (n.d.). *Turn your data into immediate impact*. <https://powerbi.microsoft.com/en-us/>
- Pukala, R., Hlibko, S., Vnukova, N., & Korvat, O. (2020). Power BI in ICT for monitoring of insurance activity based on indicators of insurance portfolios. In 2020 *IEEE International Conference on Problems of Infocommunications. Science and Technology (PIC S&T)* (pp. 393–401). IEEE. <https://doi.org/10.1109/PICST51311.2020.9467993>
- Qorri, A., Mujkić, Z., & Kraslawski, A. (2018). A conceptual framework for measuring sustainability performance of supply chains. *Journal of Cleaner Production*, 189, 570–584. <https://doi.org/10.1016/j.jclepro.2018.04.073>
- Rahman, M. H., & Rahman, A. (2019). Strategic fit: Model development and fitness analysis of a manufacturing unit. *Production and Manufacturing Research*, 7(1), 44–66. <https://doi.org/10.1080/21693277.2019.1583143>

- Rahman, M. H., & Rahman, A. (2020). Strategic fit strategy formulation: Keys to enhancing competitiveness and improving capabilities of a manufacturing unit. *Production and Manufacturing Research*, 8(1), 59–79. <https://doi.org/10.1080/21693277.2020.1742234>
- Richards, G., Yeoh, W., Chong, A. Y. L., & Popovič, A. (2019). Business intelligence effectiveness and corporate performance management: An empirical analysis. *Journal of Computer Information Systems*, 59(2), 188–196. <https://doi.org/10.1080/08874417.2017.1334244>
- Sein, M. K., Henfridsson, O., Purao, S., Rossi, M., & Lindgren, R. (2011). Action design research. *MIS Quarterly*, 35(1), 37–56. <https://doi.org/10.2307/23043488>
- Sellitto, M. A., Pereira, G. M., Borchardt, M., Da Silva, R. I., & Viegas, C. V. (2015). A SCOR-based model for supply chain performance measurement: Application in the footwear industry. *International Journal of Production Research*, 53(16), 4917–4926. <https://doi.org/10.1080/00207543.2015.1005251>
- Sherer, S. A. (2014). Advocating for action design research on IT value creation in healthcare. *Journal of the Association for Information Systems*, 15(12), 860–878. <https://doi.org/10.17705/1jais.00384>
- Tim, Y., Chiew, T. K., Lim, H. M., Teo, C. H., & Ng, C. J. (2023). Design process knowledge for crisis-driven information systems solutions: Insights on building digital resilience from an action design research study. *Information Systems Journal*. <https://doi.org/10.1111/isj.12457>
- Trieu, V. H. (2017). Getting value from business intelligence systems: A review and research agenda. *Decision Support Systems*, 93, 111–124. <https://doi.org/10.1016/j.dss.2016.09.019>
- Vuong, H. T., Jr. (2020). Monitoring and managing KPIs of material handling performance in operative logistics. <https://urn.fi/URN:NBN:fi:amk-2020052012725>
- Wang, Z., Yan, R., Hollister, K., & Xing, R. (2009). A relative comparison of leading supply chain management software packages. *International Journal of Information Systems and Supply Chain Management (IJISSCM)*, 2(1), 81–96. <https://doi.org/10.4018/jisscm.2009010106>
- Widjaja, S., & Mauritsius, T. (2019). The development of performance dashboard visualization with power BI as platform. *International Journal of Mechanical Engineering and Technology*, 10(5), 235–249.
- Wing, J., Andrew, T., & Petkov, D. (2017). Choosing action design research for the process of development, application and evaluation of a framework. In *2017 1st International Conference on Next Generation Computing Applications (NextComp)* (pp. 135–140). IEEE. <https://doi.org/10.1109/NEXTCOMP.2017.8016188>
- Zaitsev, A., & Mankinen, S. (2022). Designing financial education applications for development: Applying action design research in Cambodian countryside. *European Journal of Information Systems*, 31(1), 91–111. <https://doi.org/10.1080/0960085X.2021.1978341>